

Phase 1 Capstone Project Report

Ames Housing: Exploratory Data Analysis & Statistical Reasoning

1. The Question

Which property characteristics are most strongly associated with house sale prices? Supporting questions: do higher-quality houses sell for different prices, do prices differ across neighborhoods, and does living area tell the same price story at different quality levels?

2. The Data and Cleaning

Item	Result
Original dataset	1,460 rows x 11 columns
Final analytical dataset	1,460 rows x 10 columns (9 original + derived QualityGroup)
Duplicates	0
Columns removed	Id and LotFrontage
LotFrontage missing	248 values (16.99%)
YearBuilt validity	1900-2010; no zero, negative or future years

Id was removed because it only labels each row. **LotFrontage** was removed instead of filled in. It was missing for 248 houses, and it was not needed for the main questions. The observed average was 70.02 ft with a spread (standard deviation) of 21.90 ft. Filling the gaps with the average would shrink the spread to 19.95 ft, and filling with the median (69 ft) would shrink it to 19.96 ft. Imputation would therefore change the variable's shape and add guesses for a column the analysis does not need.

Outliers. The IQR rule (upper bound \$410,550) flagged six sale prices: Id 320 (\$595,000), 811 (\$429,800), 845 (\$412,300), 1,153 (\$412,200), 1,433 (\$420,000) and 1,444 (\$422,400). Each was checked against living area, quality and neighborhood. Five are large and/or high-quality houses in expensive neighborhoods (NridgHt, NoRidge). Id 320 (MeadowV) stands out because its quality rating is only 4 for a \$595,000 price. It deserves attention, but nothing in the data proves an entry error, so all six were kept.

3. What We Found

Sale prices are right-skewed: most houses are mid-priced and a few are very expensive. Larger living areas tend to sell for more, but the link is not perfect. Neighborhoods differ a lot in typical price. Overall quality has the strongest link to price ($r = 0.80$), followed by living area ($r = 0.60$).

Notable finding: living area alone does not explain price. At similar living areas, higher-quality houses sit in higher price ranges. This was carried into the hypothesis tests.

4. Statistical Tests

Analysis	Result	Meaning
Welch t-test (quality 7+ vs 5 or below)	$t = 36.33$; $p = 4.75 \times 10^{-149}$	Average prices differ strongly
Cohen's d	2.50	Very large effect
Sensitivity check (Quality 6 included with high)	$t = 31.72$; $d = 1.73$	Smaller effect, same conclusion
ANOVA (15 neighborhoods)	$F = 400.10$; p is about 0	Neighborhood averages are not all equal
Eta squared	0.795	Very large association
Levene's test	$p = 0.098$	No strong sign of unequal variances

Quality groups. The main test compares High Quality (rating 7 or more) with Low/Medium Quality (5 or less). Quality 6 (293 houses) stays in the cleaned data but is left out of this test to compare clearly separated groups. This can make the gap look bigger. To check, the test was repeated with Quality 6 included: the effect fell from $d = 2.50$ to $d = 1.73$, but high-quality houses still sell for far more. This exclusion is a stated limitation.

Assumptions. Sale price is not normally distributed. Shapiro-Wilk tests reject strict normality for both quality groups and for 2 of the 15 neighborhoods (MeadowV and Timber). Because the groups are fairly large, Welch's

t-test and ANOVA are kept as robust methods, and the non-normality is acknowledged as a limitation. Levene's test was not significant, so equal spread across neighborhoods is reasonable.

5. What It Means

Overall quality, living area and neighborhood are all strongly associated with sale price in this dataset. Tukey HSD was used after ANOVA to find which neighborhood pairs differ. Effect sizes are reported next to p-values so that statistical significance is not confused with practical importance. These are associations, not proof of cause and effect.

6. Limitations

The data are observational, so causal claims are not justified. Some neighborhoods have fewer houses than others. LotFrontage was removed rather than filled in, so nothing here says how it relates to price. The main quality test leaves out Quality 6, and sale price is skewed, so the results are evidence from this dataset rather than universal rules for real estate.

Conclusion: the project follows a full analysis workflow: ask a question, check and clean the data, explore patterns, investigate a notable finding, test hypotheses, check assumptions, measure effect sizes and report the evidence clearly.